



Groundwater storage in Quaternary deposits and bedrock buffers the discharge response of a small alpine catchment to climate change

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Abstract

Alpine areas play a major role in the water supply of downstream valleys by releasing water during dry periods. Groundwater storage could significantly influence how changing climatic conditions affect the discharge regime. However, these processes are still not well understood in alpine areas. Studies on how climate change impacts water resources in alpine areas rarely consider the influence of geology explicitly. An integrated surface–subsurface hydrologic model (HydroGeoSphere) was used to highlight the effect of groundwater in buffering future summer low flows and to give new insights into the role of geology in controlling discharge. The spatially explicit model allows the quantification of groundwater storage and its variation in different geological formations of an alpine catchment, under current and future conditions. The results suggest that, under future extreme climate change conditions, the average catchment groundwater storage in Quaternary deposits increases in winter and decreases in summer; the annual mean groundwater storage for the entire catchment decreases due to the decrease in mean annual groundwater recharge; and the total catchment discharge decreases and low-flow period shifts from winter to late summer by the end of this century. Future summer low flows will remain higher than the current winter low flows but will approach them closely after very dry summers. Proportionally, the decrease in groundwater storage will be less severe than the decrease in discharge. Finally, a sensitivity analysis illustrates that both Quaternary deposits (especially moraine and talus units) and bedrock play an important role in supporting discharge during low-flow periods.

Keywords Mountain hydrology · Mountain hydrogeology · Surface–subsurface flow model · Groundwater storage · Climate change

Introduction

Alpine areas are defined as mountainous regions with steep slopes located mainly above the tree line. These regions sustain water resources in downstream valleys by releasing meltwater during warm and dry periods. This seasonal redistribution can significantly reduce the implications of meteorological droughts in lowlands during summer (Beniston and Stoffel 2014; Rohrer et al. 2013). However, the reduction of snow cover due to climate change can affect the sustainability of these water reservoirs (Barnett et al. 2005; Viviroli et al. 2007).

In snow-dominated areas, rising temperatures lead to shorter snow cover durations (Klein et al. 2016), earlier melt onset and higher proportions of rain instead of snow during winter (Perico et al. 2022; Harpold and Brooks 2018; Li et al. 2017; Zhang et al. 2015) leading to a decrease in snow accumulation. For instance, Marty et al. (2017) show that, in the Swiss Alps, snow storage may decrease by 50% at

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elevations above 3000 m above sea level (asl) and almost no snow may accumulate at elevations lower than 1200 m asl by the end of the century. In response, streamflows are expected to increase in winter (Laaha et al. 2016) and decrease in spring and summer (Barnhart et al. 2016; Teutschbein et al. 2015) leading to reduced annual high flows occurring earlier in the year (Arnoux et al. 2021; Halloran et al. 2023). These trends have already been observed in the pre-alpine areas where Birsan et al. (2005) found increases in winter flows. For pre-alpine catchments, monthly streamflow in March has already significantly increased over the past decades, while monthly streamflow in May has decreased (Stähli et al. 2021). These projected changes will reduce discharge at the end of summer and autumn (Barnett et al. 2005; Lauber et al. 2014), which can considerably affect water management in downstream valleys.

Although groundwater is often under-represented in mountain hydrology (Juen et al. 2007; Huss et al. 2008; Lauber et al. 2014; Garvelmann et al. 2017; Jódar et al. 2016; Staudinger et al. 2017; Zuecco et al. 2019), recent studies show it is a major contributor to discharge (Clow et al. 2003; Liu et al. 2004; Huth et al. 2004; Hood et al. 2006; Cras et al. 2007; Hood and Hayashi 2015; Jódar et al. 2017; Glas et al. 2019; Saberi et al. 2019; Somers et al. 2019; Thornton 2024; van Tiel et al. 2024; Chartier-Rescan et al. 2025; Weiler et al. 2025). A few recent papers (Halloran et al. 2023; He and Hayashi 2023; Somers and McKenzie 2020; Arnoux et al. 2020; Hayashi 2019; Cochand et al. 2019b) highlighted the critical role of heterogeneous aquifers, located mainly in unconsolidated Quaternary deposits (such as talus and moraines) in storing groundwater in alpine areas. Nevertheless, hydrogeological processes such as surface–subsurface interactions and groundwater flow dynamics remain insufficiently studied (Halloran et al. 2026). To date, it remains unclear in which geological units groundwater is mainly stored and to what extent groundwater stored in alpine catchments can mitigate the effects of climate change (Hayashi 2019).

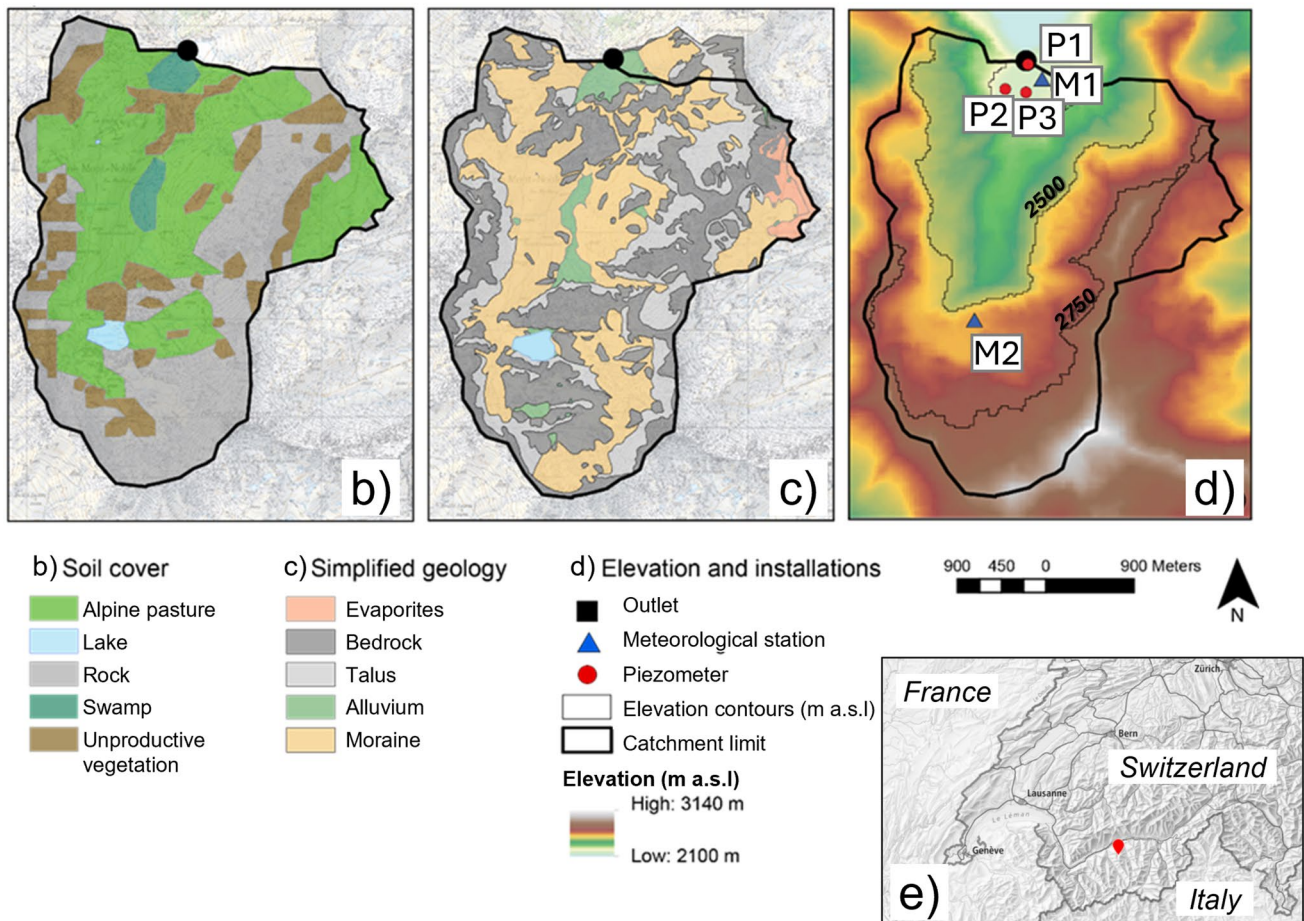
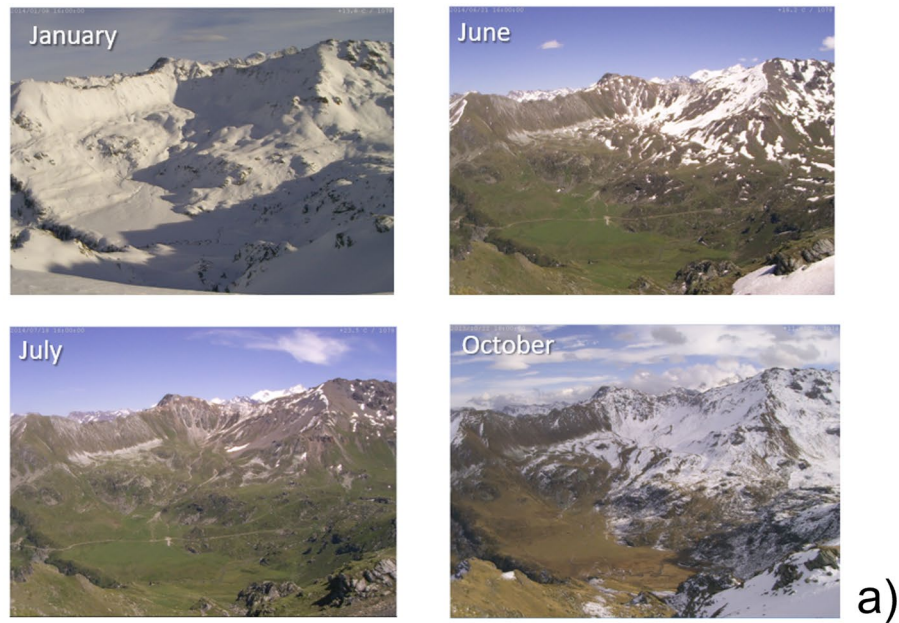
Hydrogeological processes are often oversimplified in hydrological models because of the complexity of alpine aquifers, complex coupling with cryosphere processes, computational constraints, and the lack of reliable groundwater observations in these remote areas (Somers et al. 2019; Thornton et al. 2022). Few integrated surface–subsurface hydrological models have been implemented in snow-dominated alpine catchments (Abhervé et al. 2025; Thornton et al. 2022) but mostly in glacierized catchments (Saberi et al. 2019; Somers et al. 2019; He et al. 2022). However, it remains difficult to extrapolate these results to future climate scenarios, and it is urgent to improve the understanding of hydrogeological dynamics and storage in alpine areas.

This study quantifies groundwater storage dynamics in snow-dominated alpine catchments using an integrated surface–subsurface hydrological model, with the goal of clarifying how distinct geological units contribute to low-flow generation under current and future climate conditions. The role of different geological units in groundwater storage has been investigated and their influence on future low flows has been assessed. The model setup, while inspired by an existing high-alpine catchment, is designed as a simplified yet process-based framework that captures the dominant hydrological and hydrogeological dynamics typical of such environments. Rather than predicting the future of a specific site, the goal is to enhance understanding of flow dynamics and storage mechanisms in complex alpine settings with interacting aquifers. Model robustness is explored through sensitivity analysis, particularly regarding uncertain subsurface parameters. Finally, the model is driven with two RCP 8.5 climate scenarios (1981–2099) to evaluate seasonal and interannual changes. This research contributes to advancing the understanding of groundwater storage dynamics and their role in regulating alpine streamflow under changing climatic conditions.

Materials and methods

Environmental setting

The geological conditions found in a typical small alpine catchment, the Vallon de Réchy, were used to develop the modelling framework. This 11 km² catchment ranges from 2150 to 3148 m asl with a mean elevation of 2552 m asl. It is part of the Pennine domain in the southeastern Swiss Alps (Valais, Fig. 1b). Mainly quartzite, gneiss, and calcschist form the rock outcrops and evaporitic rocks (gypsum and cargneule) outcrop in the southeastern part of the catchment (Challandes 1992). Major normal east–west faults combined with glacial erosion have created a step-like landscape consisting of a sequence of rock basins and riegels, as commonly encountered in alpine glacial valleys (Marthaler et al. 2008). A small lake (Lake Louché, 0.047 km²) is located on the uppermost plateau. The two lower rock basins are filled with moraine, alluvial deposits and peat (Fig. 1b; Cochand et al. 2019b). Talus deposits, moraine, alluvial fans, and sediments of relict rock glaciers partly cover the slopes of the valleys and two small active rock glaciers are present in the top part of the watershed. Permafrost might exist along the ridges at elevations above 2700 m asl (Lugon and Delaloye 2001) but its influence on hydrological flowpaths and groundwater recharge is considered negligible in this study, given its limited spatial extent. Similarly, potential effects of tectonic activity on groundwater resources, which are sometimes assumed to



influence water springs in mountainous areas (Barfal et al. 2022), are not considered within the scope of this study. The catchment geology is described in detail in Cochand

et al. (2019b) with a geological map and a cross section. The geological map is also available in the electronic supplementary material (ESM).

Fig. 1 **a)** Photographs (2014) of the north-eastern section of the catchment (primarily showing the lower part of the catchment); **b)** simplified soil cover of the study site (GEOSTAT 2007); **c)** simplified geology based on Swiss Geological Atlas 1:25000/Vissoie (Escher et al. 2008) and **d)** location of the discharge monitoring station (black dot), piezometers (P1, P2, and P3) and automated weather stations (M1 and M2), the grey contours represent the 250 m elevation bands used for the precipitation and temperature gradients in the model (the topographic map has a horizontal resolution of 25 m and is based on the digital elevation model swissALTI3D; Wiederkehr and Möri 2013); **e)** Map of Switzerland with the study site in the Valais region (red dot). Photograph accreditation: Marion Cochand

The discharge is snowmelt-dominated, with a snow cover period ranging from November to May–June. It has no glacier cover and precipitation averaged around 1100 mm/year during the observational period (2013 to 2018). The catchment is located almost entirely above the tree line (Cochand et al. 2019b). The soil cover is mainly located in the bottom part (Cochand et al. 2019b; Fig. 1a). The work of Cochand et al. (2019b), focusing on groundwater storage and its contribution to discharge in the Réchy catchment, has improved the hydrogeological knowledge of this region and showed that the hydrogeological catchment area differs from the topographic catchment (Fig. 1c). The evaporites zone, located in the southeastern part of the catchment, has been identified by hydrograph separation as an important aquifer for the catchment storage. Cochand et al. (2019b) also found that the dynamic groundwater storage, i.e. the annual difference between the highest and lowest observed groundwater storage, was around 300 ± 60 mm in 2013. The snow water equivalent (SWE) measured by light detection and ranging (LiDAR) in winter 2012/2013 during the period of maximum snow accumulation period was estimated at 660 ± 32 mm (Cochand et al. 2019b).

Data collection and processing

Meteorological parameters were recorded at two stations from April 2014 to 2020, one at the Ar du Tsan plateau automatic weather station (M1, 2193 m asl) and one at Lake Louché (M2, 2567 m asl), as illustrated in Fig. 1c. Liquid and solid precipitation were collected with an OTT Pluvio² and then corrected for wind undercatch. As already estimated in the previous study of Cochand et al. (2019b), a correction factor for precipitation of 3% is used for liquid precipitation (Larson and Peck 1974; WMO 2008; Cochand et al. 2019b). Evaporation and splash-out losses are assumed to be negligible (Dingman 2015; WMO 2008). The stream water level is measured hourly at the outlet of the catchment since April 2014 with two pressure probes (the second one is for back-up). The corresponding rating curves are established on the basis of 28 discharge measurements for one probe and 22 for the second probe. The discharge rate

is measured at least twice a year since 2014, using the salt dilution method, covering a wide range of discharge values, from 0.1 to 1.5 m³/s. The groundwater levels are measured hourly with pressure sensors in three piezometers (P1, P2, and P3; Fig. 1c). These piezometers are located in the alluvial plain of Ar du Tsan and range in depth from 3.8 m (P2 and P3) to 4.4 m (P1).

Mean daily snow water equivalent (SWE) and snow water input (SWI, i.e. rain and snowmelt) comes from a physically based snowpack model, run at 250 m spatial resolution and hourly temporal resolution across Switzerland for 3 years (period 01 September 2015 to 31 August 2018). Daily snow depth observations from over 300 Swiss snow monitoring stations are assimilated into this snow model. Details about these snow monitoring data, snowpack modelling, and data assimilation methods are available from Griessinger et al. (2019) and Winstral et al. (2019). The resulting 250 m gridded product is then upscaled to obtain daily, catchment-averaged SWE values that are used to calibrate the snowmelt parameters of the hydrogeological model.

Integrated surface–subsurface flow model

Model description

HydroGeoSphere (HGS; Aquanty 2020) is an integrated surface–subsurface flow model that simulates spatially distributed hydro(geo)logical processes, interactions and feedback mechanisms (Brunner and Simmons 2012). Topography, surface runoff areas, soil units, aquifers and aquicludes are physically represented. HGS, using 3D physically based equations, simultaneously simulates surface and subsurface flows, their interactions (by infiltration or exfiltration), and the key hydrological variables, evapotranspiration and discharge at the outlet of the catchment. HGS employs a control-volume finite element approach to solve Richards' equation for 3D variably saturated subsurface flow coupled with the 2D depth-averaged surface flow equation of Saint–Venant for the diffusion-wave approximation (Goderniaux et al. 2009 and Cochand et al. 2019a). The hydraulic head, the degree of saturation, and the water Darcy flux are calculated at each node. The water depth (height of water above the ground surface) and the fluid flux are computed for each surface node (Goderniaux et al. 2009). The surface nodes with water depth greater than zero form the streams. Rainfall, snowfall and potential evapotranspiration (PET) are computed at the model surface as boundary conditions. Snowfall is estimated using a simple air temperature threshold below which precipitation on a given day falls as snow (see Appendix 1 for parameter values). Snowmelt is calculated with a degree-day approach (Hock 2003), i.e. as a linear function of air temperature if this air temperature exceeds a critical threshold. In the formulation used here (Cochand et al. 2019a), this critical threshold is taken equal

to the snowfall accumulation threshold. To solve the nonlinear equations, HGS uses the iterative Newton–Raphson solver. To estimate actual evapotranspiration (AET) from PET, HGS uses Kristensen's and Jensen's (1975) model: AET is calculated from PET, soil water saturation, leaf cover, root depth, canopy rainfall interception, and the evaporation depth limit (Cochand et al. 2019a). HGS calculates water partitioning between surface runoff, infiltration and AET. Information on the underlying blueprint of the HGS model can be found in Simmons et al. (2020), while further details on the equations solved are provided in Schilling et al. (2019) and Cochand et al. (2019a). The spatially explicit simulation of interactions between rivers and aquifers is of particular interest here because water exchange fluxes, especially groundwater recharge, are crucial for the projections that are the focus of this study. HGS has been used successfully in combination with climate change scenarios in Europe (Goderniaux et al. 2009) and in Canada, with a particular focus on snow-related processes (Cochand et al. 2019a).

Setup of the base model

Geology Due to limited data, the subsurface was simplified using the detailed hydrogeological assessment of Cochand et al. 2019b. The surface geology is thus attributed to the surface mesh based on the simplified geological map (Fig. 1b), as illustrated in Fig. 2. The primary geologic components of the catchment are bedrock, evaporites and unconsolidated Quaternary deposits (QD). QD consists mainly of three different hydrofacies: alluvium, talus and moraines, with the distribution shown in Fig. 1b. The soil cover, composed of alpine pasture, is located mainly in the bottom part, as illustrated in Fig. 2. The thickness of sediments in the flat, low-lying areas of the catchment was determined using geophysical methods and sediment cores (Geotest 1963), indicating a maximum depth of approximately 20 m. In this flat area, slopes range between 0 and 5°. On the basis of the geomorphology and field observations, QD depth was assumed to decrease with increasing slope. Following that, the depth of the QD in the model is built as follows: depth is equal to 10 m for slopes between 5 and 10°, 8 m for slopes between 10 and 15°, 5 m for slopes between 15 and 25°, 3 m for slopes between 25 and 30°, 2 m for slopes between 30° and 35° and 1 m for slopes above 35° (Fig. 2). The rest of the model domain is composed of bedrock. Bedrock represents an important volume in the model due to the important elevation range of the catchment and the relatively low depth of QD. It is important to emphasize that the retained parameterization of the geological structures represents the main features of the study domain, relevant to simulate dominant hydrological and hydrogeological processes rather than a fully validated reconstruction of the subsurface. Consequently, the analysis focuses on storage changes (i.e. dynamic groundwater storage and mean trends),

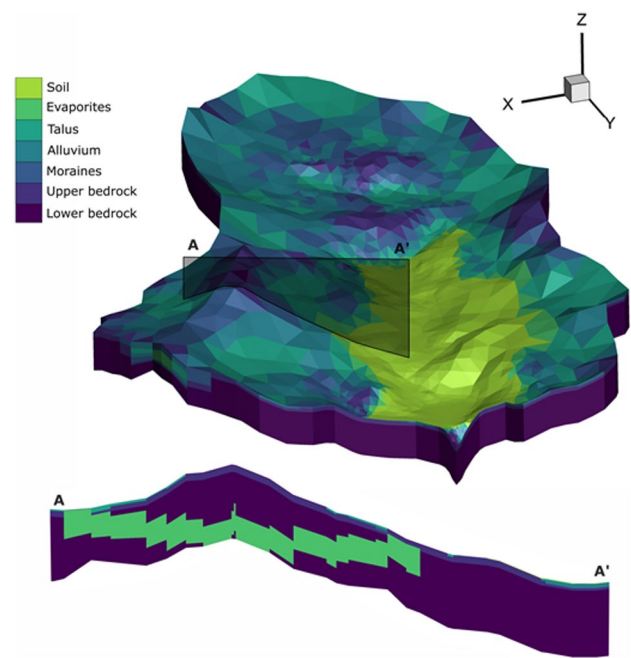


Fig. 2 Illustration of the HGS model and geology

rather than on absolute water storage. The relative contributions of different geological units to catchment-scale discharge are therefore analysed.

Mesh The software AlgoMesh (version 1.2.1, 2016) is used to build a 3D finite-element mesh for the catchment. The surface flow domain is first generated using a 2D horizontal triangular element mesh using the method described in Käser et al. (2014) with refinement near rivers to capture steep topographic gradients. The generated 2D mesh contains 1421 nodes and 2714 triangular elements. Similar to the setup described by Cochand et al. (2019a), the 3D triangular prism elements mesh is then generated by replicating the 2D mesh along the vertical. The resulting 3D mesh has 29 element layers and 29 nodal slices over a total vertical thickness of 285 m, with 41,209 nodes and 75,992 triangular prism elements in total. The elevation of the top of the mesh (i.e. the ground surface) is obtained from the 20 m digital elevation model (Fig. 1c). The vertical discretization of the near-surface layers is set to a finer resolution close to the ground surface because it plays a crucial role in the simulation of surface–subsurface interactions and unsaturated flow (Cochand et al. 2019a). Therefore, the top 2 m from the ground surface is discretized into four layers. The thicknesses of the top four layers increase gradually downward with values of 2.5, 24.5, 49 and 124 cm. Below these, four layers with a constant thickness of 2 m extend to a depth of 10 m. These are followed by five layers with a constant thickness of

10 m, reaching 60 m depth. Finally, 15 additional layers, each 15 m thick, extend down to 285 m.

Boundary conditions The lateral boundaries are assigned as no-flow boundary conditions for groundwater flow and similarly, the bottom boundary is assumed to be impermeable. This last assumption can be justified by the fact that groundwater flow in the deeper bedrock is likely negligible because of its very low hydraulic conductivity. Finally, a critical-depth boundary condition is implemented at the catchment outlet to allow both surface and subsurface water to flow out of the model. This boundary condition represents the only outflow location in the model.

Three different flux boundary conditions are assigned at the top of the mesh: rainfall, snowfall, and PET. Daily PET is computed from temperature and latitude, following the modified McGuinness equation (Oudin et al. 2005). Elevation is segmented into four bands of 250 m (Fig. 1c), and gradients of 3% per 100 m for precipitation and of 0.3 °C per 100 m for temperature are applied. These gradients are determined from the elevation lapse rates calculated from the two meteorological stations within the watershed (Fig. 1c). The volume of water pumped in the river for drinking water supply (leaving the catchment) is measured continuously and has been added to the discharge to consider the total outflow of the catchment.

Parameters The hydrogeological parameters of the model (hydraulic conductivity and porosity) are manually adjusted to reproduce the dynamics of both hydraulic heads from the three piezometers and the discharge at the catchment outlet with a specific target on low-flow periods to ensure that the model represents both surface and subsurface dynamics. The objective is not to reproduce the water levels and discharge precisely, but rather to capture their main variations to explore the dynamics depending on the aquifer units, their contribution to discharge and to discuss how such areas might evolve with climate change.

The first step of the manual setup of the base model is a 10-year steady-state run using the average values for precipitation, temperature and PET. The aim is to reproduce the average discharge and to equilibrate initial conditions. For the second step, the model is run in transient conditions, starting with the final simulation outputs of the steady state run as initial conditions. The parameters of the surface–subsurface model are adjusted on the basis of a 3-year period (01 September 2015 to 01 September 2018) and the simulations are checked on another 1.5-year period (21 October 2014 to 01 April 2016).

Hydrogeological parameters (porosity and hydraulic conductivity) are set within their range of possible values from the literature (Vincent et al. 2019) and then manually adjusted (Appendix 1). The other parameters (such as specific storage, unsaturated van Genuchten parameters, etc.) are defined on the basis of standard values found in the

literature (see the ESM for the other parameter values). The model water inputs are the daily SWI output from the mass- and energy-balance snow model (see the “Mean annual variations and response of groundwater storage” section). In this step, snow is not directly simulated as it is already included in the SWI modelling. In addition, PET is set to zero above 2600 m asl, as assumed by Cochand et al. (2019b).

The results of the manually adjusted model for the simulated discharge are presented in the ESM. Two metrics are used here for the base model setup: the volumetric error (E_v) and the Nash–Sutcliffe model efficiency coefficient (NSE; Nash and Sutcliffe 1970):

$$NSE = 1 - \frac{\sum (x_{obs} - x_{sim})^2}{\sum (x_{obs} - \bar{x}_{obs})^2} \quad (1)$$

$$E_v = 1 - \frac{\sum (|x_{obs} - x_{sim}|)}{\sum x_{obs}} \quad (2)$$

where x_{obs} and x_{sim} are observed and simulated SWE or discharge. The NSE metric varies between $-\infty$ and 1, with a value of 1 representing a perfect fit, while negative values indicate performance worse than using the observed mean as a predictor. Simulated and observed discharge and hydraulic heads in three piezometers are presented in the ESM. The two model performance metrics for discharge are both satisfactory: the NSE for discharge is equal to 0.63 and E_v to 4.3% (Table 1) over the 3-year period. The seasonal variations and recession levels during low-flow periods of water levels in the three piezometers are quite well reproduced although the simulated head elevation shows a positive shift. Subsequently, the calibrated hydrogeological model is run with a degree day model for snow accumulation and melt over the same period. The snow parameters are adjusted to reproduce the catchment-average daily SWE values derived from the physically based snowpack model (see the “Data collection and processing” section) during the same 3-year period. The snow parameters are snow density, the snow-melt and sublimation rates and the snow accumulation and snowmelt threshold (see the ESM for parameter values). The snowmelt rate parameter is taken from the HBV model calibrated with PEST on the same snow data in the work of

Table 1 Model performance metrics for the base model setting

Performance metric	Setup (period 01 September 2015 to 01 September 2018)		Validation (period 21 October 2014 to 01 April 2016)
	SWE	Discharge	Discharge
NSE	0.85	0.63	0.73
E_v (%)	−15.0	4.3	8.2

Arnoux et al. (2021). The snowmelt onset threshold is set to $+0.5\text{ }^{\circ}\text{C}$ (Senese et al. 2014) to fit the data. The model performance metrics for calculated SWE are both satisfactory: NSE equals 0.85 and $E_v -15.0\%$ (Table 1).

The adjusted model is validated by comparing simulated and observed streamflow over an independent 1.5-year period (Fig. 3). The two model performance metrics for discharge are both satisfactory: NSE is equal to 0.73 and E_v to 8.2%. Some discrepancies remain between observed and simulated high flows, probably due to an underestimation of SWE. The low flows are well reproduced (Fig. 3).

It is important to note that the aim was not to reproduce exact parameter values. Equifinality cannot be avoided, as it remains an inherent challenge given the current limitations in observational data. To date, insufficient observations in Alpine catchments prevent the identification of unique parameter sets. The aim of this study was to reproduce the general dynamics of the system and to understand the relative role of different geological units in buffering the discharge response. In addition, through a sensitivity analysis (see the “Groundwater storage in relation to hydraulic conductivity” section), a wide range of parameters for hydraulic conductivity in the geological settings (one order magnitude lower and higher) were generated. The sensitivity analysis does not resolve equifinality, but it allows a solid exploration of the parameter space in relation to the dynamics of storage in the different geological units.

The change of hydraulic parameters with temperatures is not considered because, in this region, the variations of groundwater temperature are minor (maximum yearly variation of $3\text{ }^{\circ}\text{C}$) and it remains cold throughout the year. Once the base model is established, it is run with climate change projections and the results of the simulations are extracted with the Tecplot software. Groundwater storage in the different geological units is obtained by integrating the water volume of the saturated zone within each unit.

Climate change projections

To simulate the impact of future changes in air temperature and precipitation on groundwater storage and discharge, the recent Swiss Climate Change Scenarios 2018 dataset was used. Following the recommendations of CH2018 (2018), seasonal variations are compared on the basis of mean values over 30 years. A reference period (1991–2020) is compared to three future periods (2035: 2020–2049, 2060: 2045–2074, and 2085: 2070–2099), as illustrated in Fig. 5. The original data set comes from global climate models (GCMs). It has then been adapted for Switzerland with downscaling and quantile mapping (CH2018, 2018) and provides daily gridded historical data (from 1981 to 2018) and projections of air temperature and precipitation (from 2018 to 2099) at 2 km resolution. From 21 available RCP 8.5 model chains, two were selected: the regional climate model SMHI-RCA4 driven by the GCMs

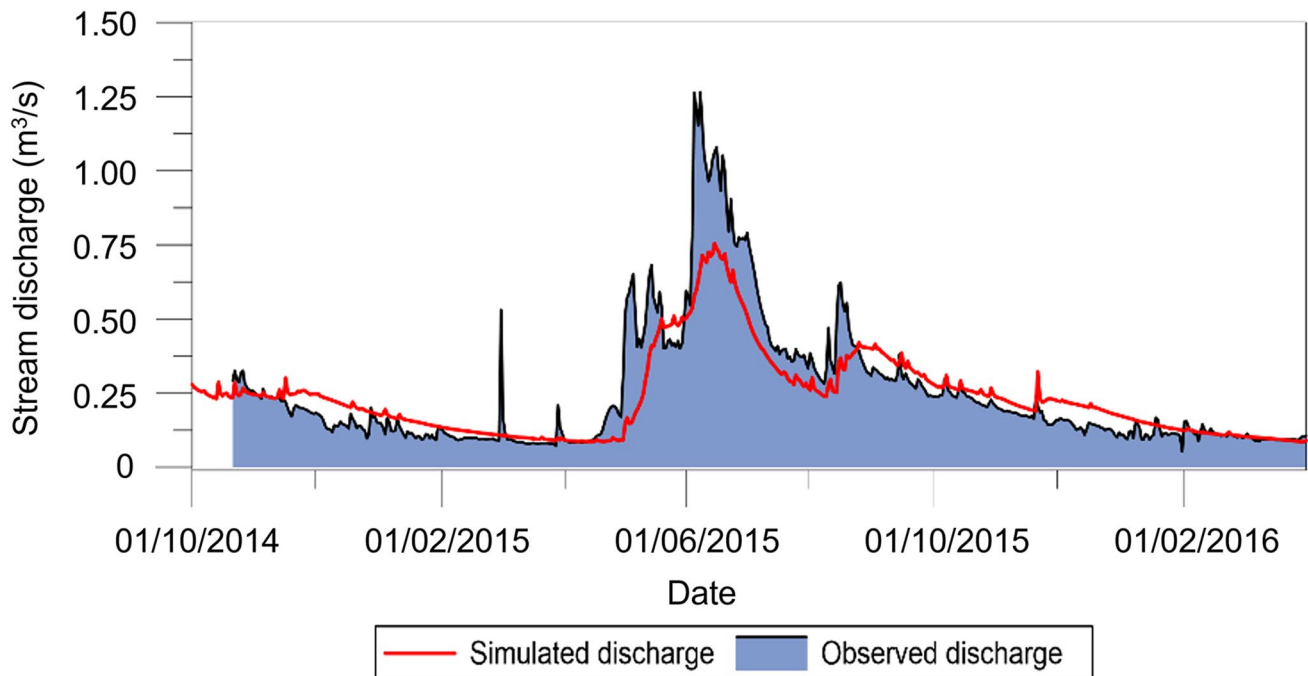


Fig. 3 Simulated daily discharge compared to the measured daily streamflow for the period 21.10.2014–01.04.2016

MPI-M-MPI-ESM-LR (S1) and CCCma-CanESM2 (S2), chosen for their relatively large spring-temperature increase compared to other model chains (CH2018, 2018). The main reason for this choice is to specifically study the effect of groundwater contribution on future summer low flows in maximizing the effect of climate change on snowmelt because high spring temperatures have a strong effect on snowmelt and thereby groundwater recharge. Therefore, the selected RCP8.5 model chains represent upper-bound climate impacts; under median warming scenarios, the simulated changes in snowmelt and its consequences would likely be less pronounced.

Catchment-scale precipitation and temperature time series were derived by area-weighting daily grid-cell values. The mean annual temperature and precipitation time series are illustrated in Fig. 4. The increase in mean annual temperature is larger for S2 than for S1 (+6.4 °C in mean for S2 and +4.8 °C for S1 from the reference to the 2085 period). Mean annual precipitation slightly decreases for S2 by around 80 mm/year, whereas with S1, mean annual precipitation slightly increases by 50 mm/year between the reference and the 2085 period (Fig. 4). Compared to scenario S1, scenario S2 presents a higher increase in spring temperatures, in addition to an important decrease in summer precipitation by 2085 and a more pronounced transient increase in spring precipitation for the near future (Fig. 5). The adjusted HGS model is run with temperature and precipitation from S1 and S2, and SWE is dynamically simulated in HGS using the degree-day model, ensuring consistent representation of snow accumulation and melt under projected climate conditions.

It is noteworthy that the effect of parameter uncertainty in the hydrological model is relatively small when compared to the uncertainties associated with future climate projections. Indeed, projections of future discharge are strongly influenced by climate model choice and internal climate variability, which generally outweigh the impact of model

parameter variations. This has been highlighted in previous studies (Evin et al. 2026), showing that uncertainties from Regional Climate Models and natural variability dominate the spread in hydrological projections. Consequently, while sensitivity to hydraulic conductivity is important for understanding current catchment dynamics, the range of potential future changes in streamflow is largely driven by climatic uncertainties rather than by the model parameters themselves.

Results and discussion

Mean annual variations and response of groundwater storage

The mean annual SWE and the corresponding annual meltwater amount follow a decreasing trend (Fig. 6b and c). This decreasing trend is slightly greater for S2 than for S1, with projected mean SWE decreases of approximately 100 mm (S1) and 120 mm (S2), and meltwater decreases of about 150 mm/yr (S1) and 245 mm/year (S2), between the reference (1991–2020) and the last simulation period (2070–2099). Mean annual SWE falls below 60 mm in both scenarios by the final simulation period, representing a substantial decrease.

The mean annual ET follows a quasi-continuous increase, resulting in a total mean increase of around 45 mm from reference to the 2085 period for both S1 and S2 (Fig. 6a). The mean annual discharge at the catchment outlet remains stable between the reference and 2085 period for S1 (−30 mm/year) while it decreases for S2 (−105 mm/year; Fig. 6d).

The mean annual groundwater storage (i.e. the water volume in all the geological units normalized by the catchment area) steadily decreases by around 110 mm (S1) and 130 mm (S2) from reference to the 2085 period (Fig. 7a). Most of the groundwater storage decrease comes from groundwater stored in

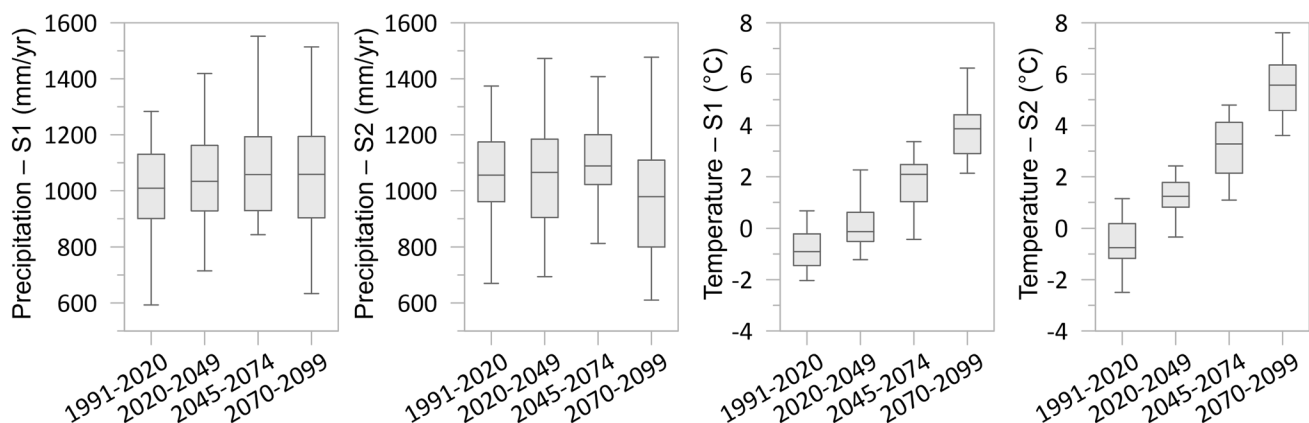


Fig. 4 Mean annual temperature and precipitation for the two-climate change RCP 8.5 model chains S1 and S2, for the reference period and the three future periods

Fig. 5 Monthly mean variation of precipitation and temperature for the two RCP8.5 projections S1 (left) and S2 (right) for the reference period and the three future periods

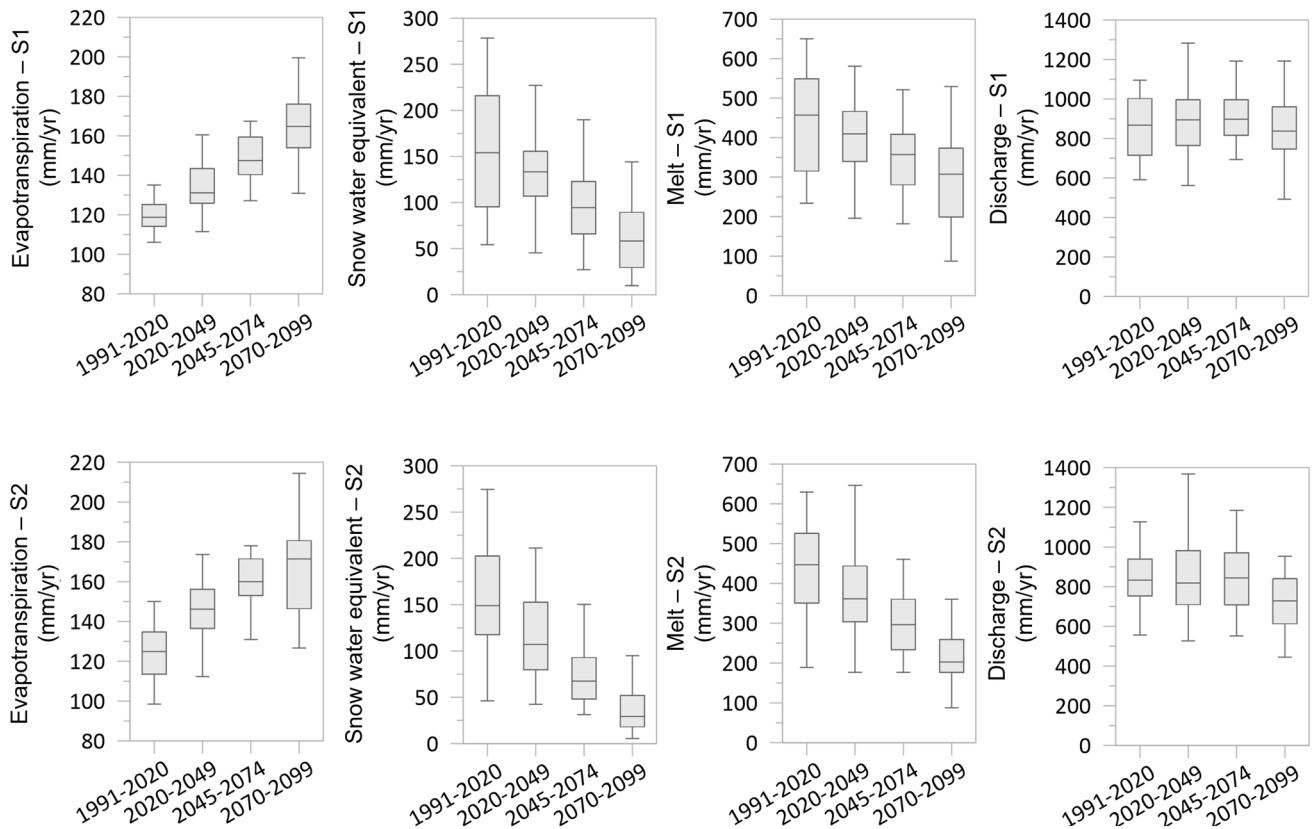
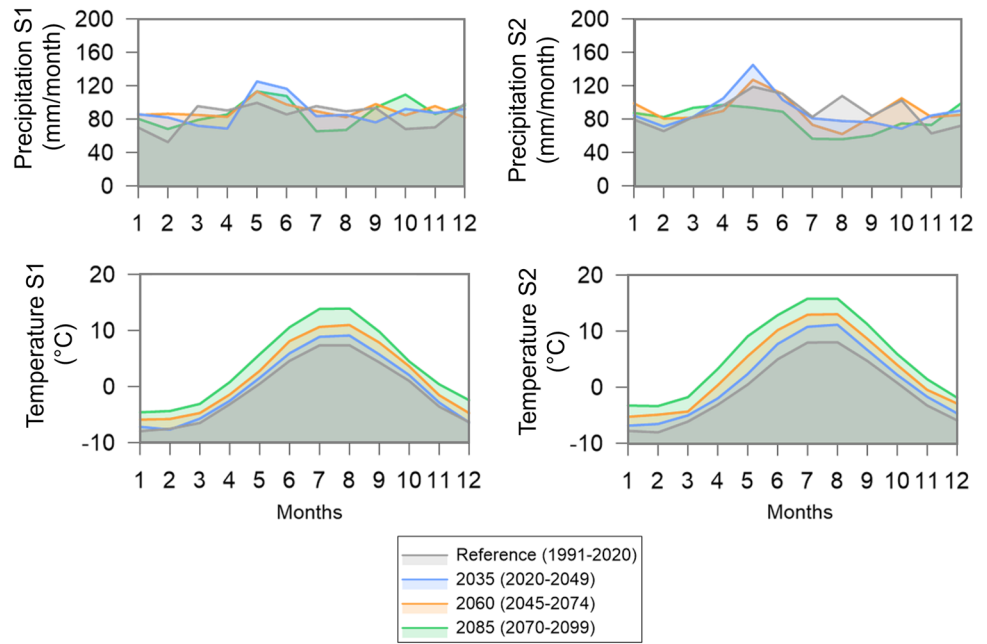


Fig. 6 Simulated dynamics of, from left to right: (a) mean annual evapotranspiration (ET), (b) snow water equivalent (SWE), (c) snowmelt (melt), and (d) discharge, following the two projections S1 (upper part) and S2 (bottom part)

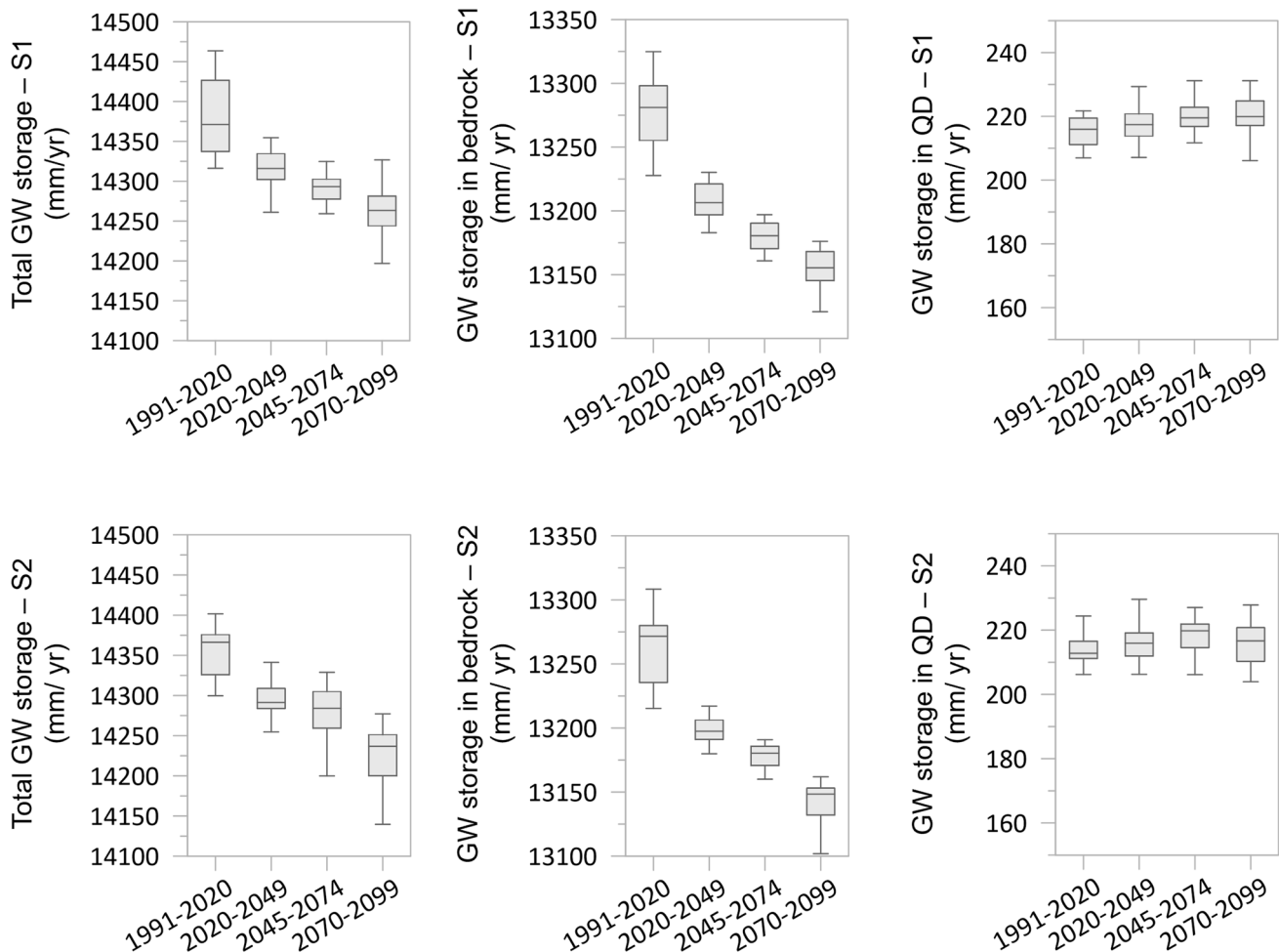


Fig. 7 From left to right: simulated mean annual total groundwater storage (a), groundwater storage in bedrock (b) and groundwater storage in Quaternary deposits (QD) (c) following both projections S1 (upper part) and S2 (lower part)

the low-permeable bedrock, which shows a decrease of around 125 mm from reference to the 2085 period (Fig. 7b); mean annual groundwater storage in QD stays stable from reference to the 2085 period (Fig. 7c). The continuous decrease of groundwater storage in bedrock is due to a decrease in mean annual recharge combined with a succession of dry years. The groundwater storage in QD follows a different trend, probably because these deposits are mainly located in the valley bottom. They are therefore recharged directly by precipitation and snowmelt, as well as by groundwater from upper bedrock (crests) and by the streams (depending on the season, see the “Mean seasonal variations and buffering capacity of groundwater” section for seasonal trends).

Mean seasonal variations and buffering capacity of groundwater

By 2085, the seasonal snowpack dynamics will strongly change. The monthly maximum SWE_{mean} (i.e. the average

catchment-scale SWE close to the time of maximum snow accumulation) will decrease from around 360 mm to 175 mm under S1 and to around 120 mm under S2 (Fig. 8). Correspondingly, the maximum SWE appears 1 month earlier (April) under S1 and 2 months earlier (March) under S2. The corresponding maximum monthly snowmelt rate is lower, with decreases of approximately 35 and 80 mm/month under S1 and S2, respectively, for the 2085 period compared to the reference period. ET increases mainly in spring due to longer snow-free periods, combined with high water availability during and after the melt. ET decreases in summer by 2085 but only under S2 because insufficient water is available in the soil unit (i.e. under S2, ET becomes water-limited).

The change in snowpack dynamics propagates to stream discharge. Monthly discharge increases in winter, with an estimated rise of approximately 20 mm in February by 2085 under both climate scenarios S1 and S2. In contrast, the snowmelt-driven discharge peak is significantly reduced, with decreases of about 23 and 70 mm/month under scenarios S1 and S2,

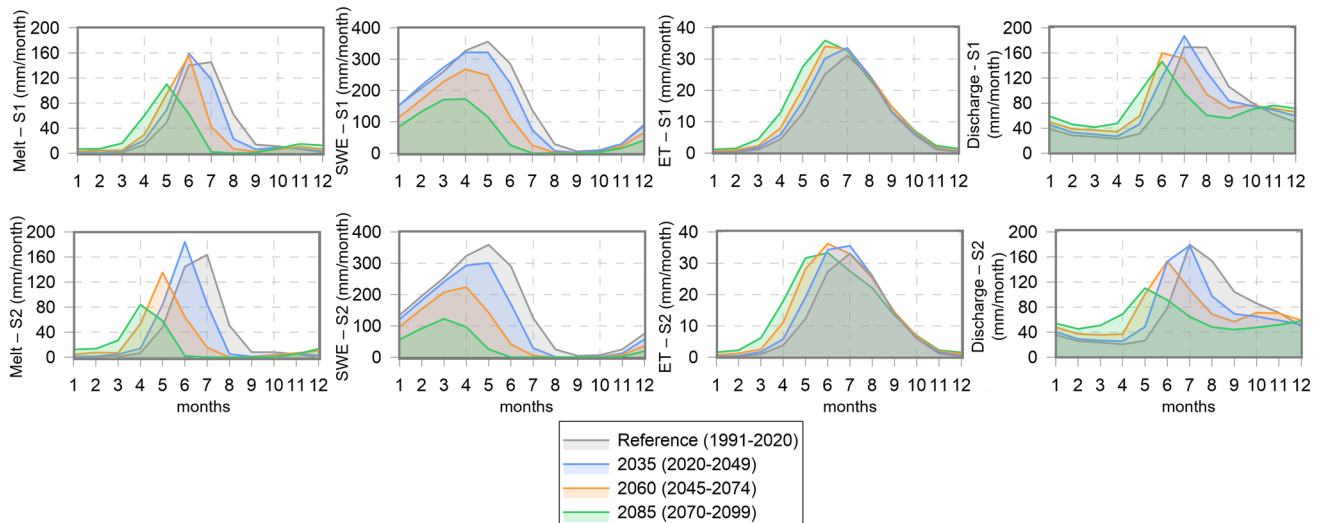


Fig. 8 Simulated dynamics of mean seasonal melt, snow water equivalent (SWE), evapotranspiration (ET) and discharge following the two climate change projections S1 and S2

respectively. Following the reduced snowmelt peak, summer discharge shows a substantial decline. Under scenario S1, discharge decreases by approximately 38% (64 mm/month) in August and 45% (48 mm/month) in September. Under scenario S2, the reductions are even more pronounced, with declines of around 45% (67 mm/month) in August and 55% (58 mm/month) in September. By 2085, discharge begins to recover after September under S1, whereas it continues to decline under S2. These diverging trends are primarily attributed to differences in the seasonal precipitation patterns between the two scenarios.

The simulated dynamic groundwater storage in QD and in bedrock indicates that seasonal discharge is mainly constituted of the QD groundwater (Fig. 9). The mean dynamic groundwater storage in QD is around 60 mm in the reference period and approximately 20 mm in the 2085 period, as illustrated in Fig. 9. In contrast, seasonal variations in bedrock storage are very small (less than 25 mm) and insufficient to support discharge alone in summer. Simulated bedrock storage shows that recharge in bedrock increases in winter and decreases in summer, but the increase is not enough to offset the summer decrease. Consequently, the mean annual recharge in bedrock decreases.

QD aquifers are recharged by surface infiltration, by stream infiltration, and likely by up flowing groundwater from bedrock where QD occupy valley bottoms. The dynamics of stream discharge and of the storage in QD are very similar as they are co-dependent. In spring, the stream recharges QD aquifers, and during low-flow periods, groundwater from QD contributes to the discharge in streams. Therefore, the groundwater buffer efficiency depends on the storage in QD.

Moreover, water storage is also sensitive to climate change: the groundwater storage in QD decreases in summer by ~11%

(28 mm) in August and by ~10% (23 mm) in September under S1, and by around ~14% (34 mm) in August and by ~12% (30 mm) in September under S2, by 2085 compared to the reference period. Therefore, in the future, summer discharge will drop by ~50% while QD groundwater storage only decreases by about 10%; groundwater resources are therefore more stable.

Future low flows shift from winter to summer

Assessing future low flows in alpine areas is essential for water management and to anticipate droughts in downstream valleys. Therefore, the simulated minimum discharges over seven consecutive days (QNM7) are presented in Fig. 10. It shows that the low-flow period starts to shift from winter to summer because of more extended low-flow periods in summer and shorter low-flow periods in winter in the future compared to the present. The period of minimum flow alternates from winter to summer between 2062 and 2091 and shifts to summer after 2092. It is also interesting to see that the annual QNM7 slightly increases from 1981 to 2099. Changes in simulated annual low flows are therefore less pronounced in future years. These evolutions highlight that future summer low flows are higher than current winter low flows, even under extreme future projections. However, in extremely dry years (for example 2092, 2093 and 2095 with S2; Fig. 10), the summer QNM7 becomes close to the current winter QNM7. These three hypothetical years in scenario S2 are characterized by particularly low summer rainfall over several consecutive months from May to September. It is also important to notice that, in the predictions used in this study,

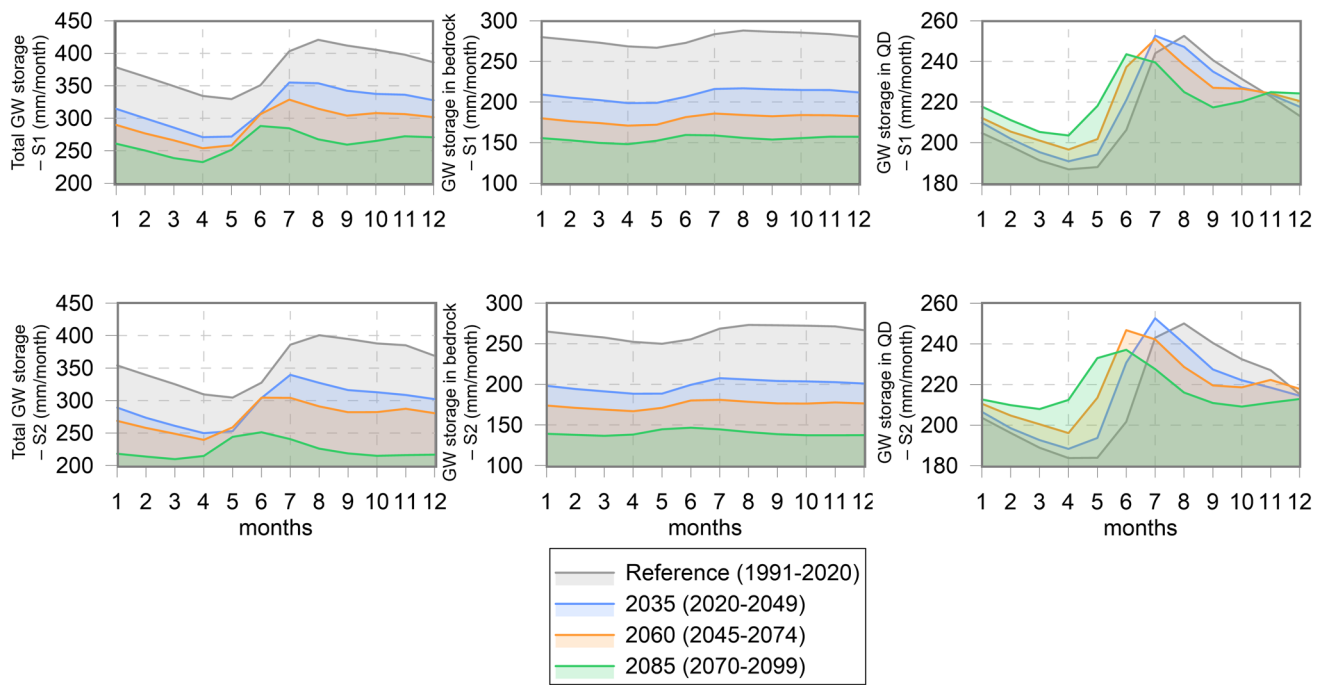
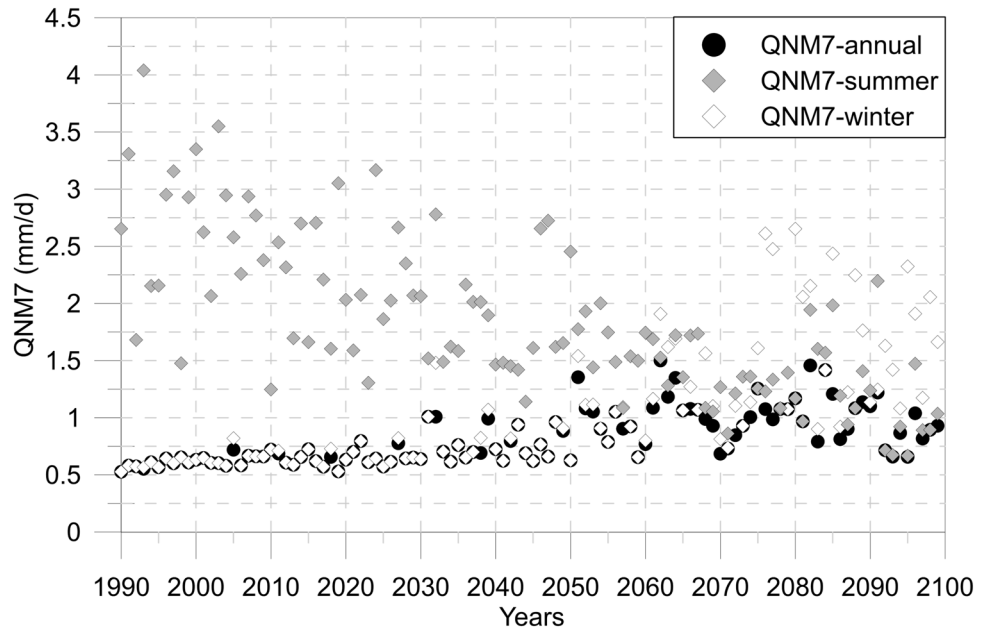


Fig. 9 Simulated dynamics of mean groundwater storage (S_{GW}) for the entire catchment (in all combined geological units), in Quaternary deposits (QD) and in bedrock, following the two climate change projections RCP 8.5 S1 and S2

Fig. 10 Simulated minimum discharge over seven consecutive days (QNM7) per year, per end of summer (August–September) and per end of winter (April–May) for S2



mean annual precipitation does not decrease. Long term precipitation patterns, however, remain challenging to predict. If precipitation decreases over several consecutive months in the future, groundwater may be able to

buffer changes in stream discharge for only limited periods, as streams are strongly dependent on groundwater contribution from QD and these aquifers react quickly to precipitation changes.

Groundwater storage in relation to hydraulic conductivity

The sensitivity analysis is conducted by varying the hydraulic conductivity of each of the five hydrogeological units independently. For each unit, the model is run twice: once with a hydraulic conductivity of one order of magnitude higher and once with one order of magnitude lower than the base model value. For example, for the moraine unit (5×10^{-3} m/s), simulations are performed with 5×10^{-2} m/s (high) and 5×10^{-4} m/s (low). This approach results in ten simulations, eight of which are presented in Fig. 11, with the corresponding storage dynamics shown in Fig. 12. To better highlight the key controls on discharge, results are described separately for high-flow and low-flow conditions.

Under high-flow conditions, increasing hydraulic conductivity in the moraine and talus units leads to higher peak discharges, whereas decreasing conductivity reduces peak

flows (Fig. 11b, c). These two units therefore exert a strong control on high flow generation. In contrast, changes in the alluvium unit have little effect on peak discharge (Fig. 11a), likely due to its limited spatial extent (Fig. 1). Bedrock properties have only a minor influence on high flows (Fig. 11d).

Under low-flow conditions, which represent the main concern for water resources, the controls differ significantly. Lower hydraulic conductivity in the moraine and talus units increases groundwater storage (Fig. 12b, c) and enhances discharge buffering, resulting in higher low flows (Fig. 11b, c). Conversely, higher conductivity reduces storage and leads to lower low flows. These effects remain highly relevant. Bedrock also plays a key role in sustaining low flows. When its hydraulic conductivity increases, groundwater storage in the deep bedrock decreases (Fig. 12d), and more water is released to streams, resulting in higher low flows (Fig. 11d). In contrast, lower bedrock conductivity increases deep storage (Fig. 12d) but does not significantly enhance discharge (Fig. 11d).

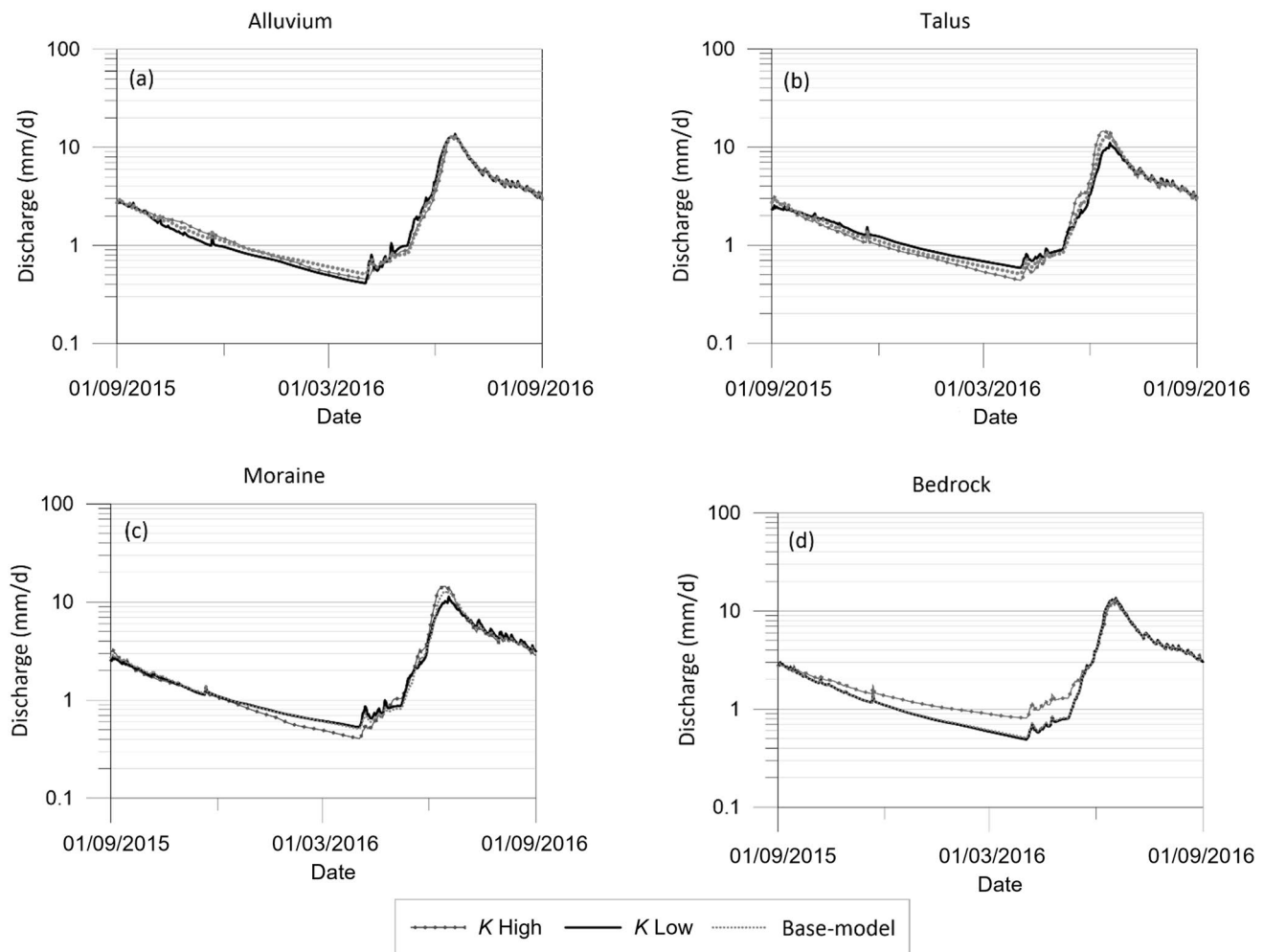


Fig. 11 Sensitivity analysis: discharge obtained by running the model with varying hydraulic conductivity of the geological units : (a) Alluvium, (b) Talus, (c) Moraine and (d) Bedrock. The base model has been run twice for each unit: once with a hydraulic conductivity of

one order of magnitude higher (K_{high}) and once with one order of magnitude lower (K_{low}) than the base model. Results are shown on a log scale to represent low flows better. The discharge is in mm/d (i.e. volumetric discharge divided by the catchment area)

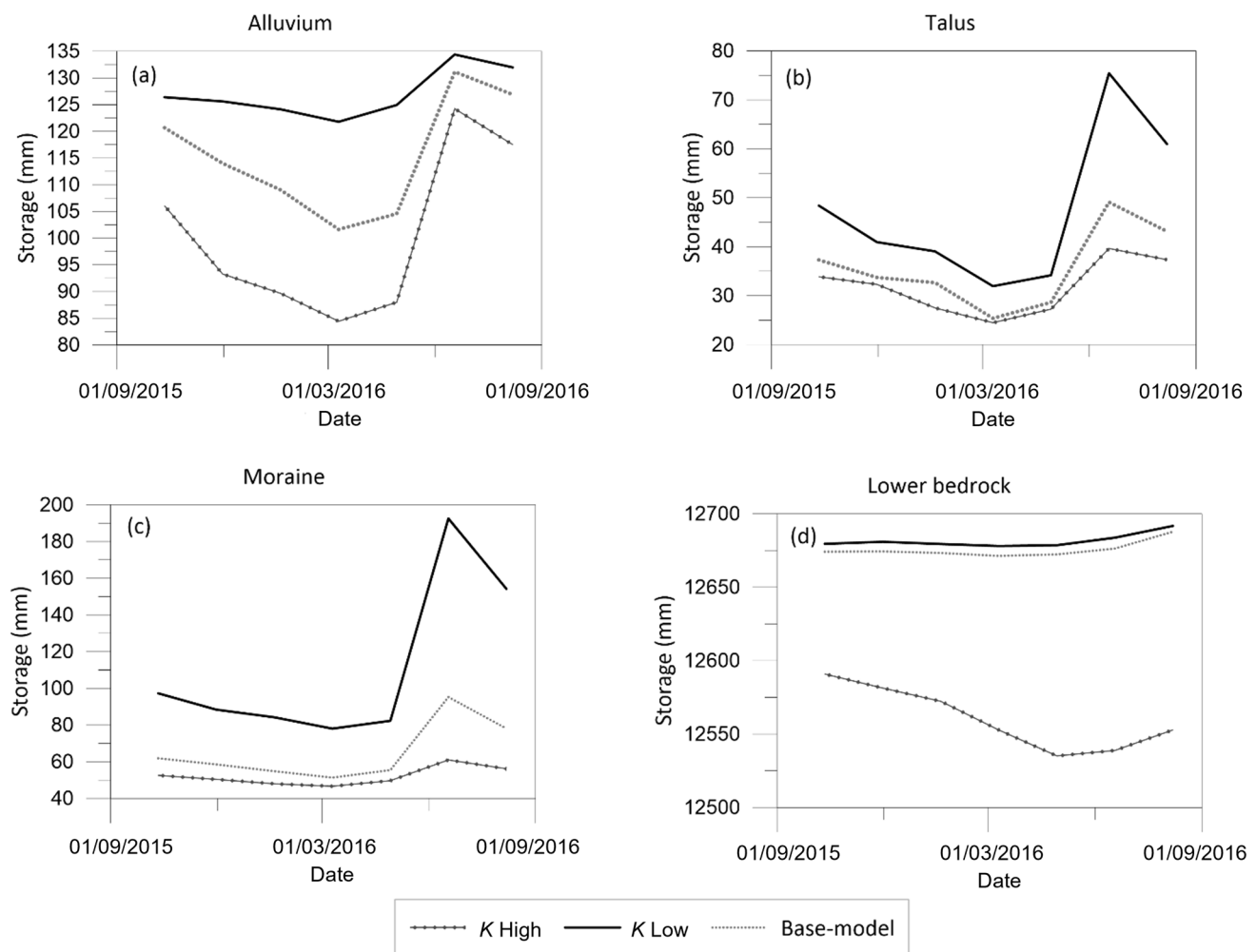


Fig. 12 Sensitivity analysis: Groundwater storage obtained by running the model with varying hydraulic conductivity of the geological units: (a) Alluvium, (b) Talus, (c) Moraine and (d) Bedrock. The base

model has been run twice for each unit: once with a hydraulic conductivity of one order of magnitude higher (K_{high}) and once with one order of magnitude lower (K_{low}) than the base model

Overall, this analysis highlights that low-flow dynamics are primarily controlled by storage and release processes in both QD (especially moraine and talus) and bedrock. When the hydraulic conductivity in the moraine and talus units is lower, the storage in both units increases, and the discharge is buffered. When the hydraulic conductivity in the bedrock unit is higher, the storage in the deep bedrock decreases, supplying more water for discharge during low flows. Thus, moraine and talus units mainly regulate buffering capacity, while bedrock controls the release of water during dry periods.

Conclusions

This paper analyses the buffering role of groundwater on future summer low flows using a modelling setup typical of small alpine catchments. Rather than providing deterministic

predictions for a specific site, the goal was to explore how a high-alpine catchment with characteristic features might respond to climate change. A base model inspired by an existing catchment was developed to account for key storage types and processes. This model was used to assess seasonal storage dynamics under current and future conditions and to perform a sensitivity analysis of hydrogeological parameters.

The novelty of this approach lies in quantifying groundwater storage in various geological units under both present and future climate conditions using the integrated HydroGeoSphere model, and in linking these storages to catchment-scale discharge.

The results highlight that, under future extreme climate conditions, mean groundwater storage in QD increases in winter and decreases in summer, while in bedrock it continuously decreases. Overall, mean annual groundwater storage decreases due to a marked reduction in summer recharge, only partially offset by higher winter recharge. By the end of the century, the annual low-flow period shifts from winter

to late summer or early autumn. Though future summer low flows generally exceed current winter low flows, they approach them after very dry summers. Discharge during low-flow periods is supported by both QD and bedrock storages, underlining their complementary roles. Because the simulations use the upper-bound RCP8.5 scenario, the magnitude of the projected changes reflects extreme conditions. It is assumed that similar processes would occur under more moderate climate scenarios, although with less pronounced effects.

Given the similarity of QD landforms across many alpine regions, these findings may provide general insights applicable to other non-glacierized alpine catchments, though site-specific variability must be considered. The results align with recent studies highlighting the critical role of heterogeneous aquifers, located mainly in unconsolidated QD, in storing groundwater in alpine areas (Cochand et al. 2019b; Glas et al. 2019; Hayashi 2019).

These insights are important for future water management, as reduced summer flows—combined with growing demand—could lead to shortages. In proportion, the decrease simulated in groundwater is less severe than in discharge. The decreases of groundwater recharge, storage and discharge, with RCP8.5 scenarios is supported by other studies in the Swiss Alps where the proportion of glacier cover is limited (Halloran et al. 2023; Arnoux et al. 2021; Muelchi et al. 2021). Such changes may

directly impact downstream users, water supply and ecosystem services, highlighting the societal relevance of groundwater dynamics under climate change.

Permafrost degradation could locally enhance recharge but is negligible in the study catchment. In glacierized catchments, melt dynamics can alter this picture, as seen in the Qinghai-Tibet Plateau (Xu et al. 2024; He et al. 2022) and the Andes (Somers et al. 2019), where groundwater temporarily buffers glacier loss but ultimately declines under increasing evapotranspiration and decreasing precipitation.

Future research should focus on improving hydrogeological data, especially in alpine areas where observations remain scarce. Longer time series and additional measurements are needed to refine models and better understand the hydrological interactions among different QD and between these deposits and the underlying bedrock.

Finally, this study highlights the need to consider geological features and groundwater dynamics in low-flow projections, a topic often overlooked in high alpine environments. While the results are exploratory, they provide a first step toward linking hydraulic properties and storage dynamics in alpine catchments. Similar model-based approaches for additional case studies with other dominant geologies and hydroclimatological conditions could yield broader insights into these critical alpine processes.

Appendix 1

Table 2 Values used for the setup of the base model for the different units

Geological units	Subsurface parameters					
	Hydraulic conductivities	Porosity	Specific storage	van Genuchten parameters		Residual saturation
	$K_{x,y,z}$ (m s^{-1})	θ (—)	S_s (m^{-1})	α (m^{-1})	β (—)	S_{wr} (—)
Talus	5×10^{-3}	0.2	1.1×10^{-5}	8.00	2.00	0.025
Alluvium	1×10^{-3}	0.2	1.1×10^{-5}	8.00	2.00	0.025
Moraine	5×10^{-3}	0.2	1.1×10^{-5}	8.00	2.00	0.025
Bedrock	1×10^{-8}	0.05	1.6×10^{-4}	9.00	2.68	0.025
Evaporites	1×10^{-4}	0.15	1.6×10^{-4}	3.60	1.56	0.025
Soil	1×10^{-4}	0.2	-	3.60	1.56	0.025

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Author contributions Marie Arnoux: Investigation, Methodology, Formal analysis, Writing—original draft, Writing—review & editing. Final review and validation of the manuscript.

Fabien Cochand: Methodology, Writing—review & editing.

Philip Brunner: Conceptualization, Supervision, review & editing, Project administration, Funding acquisition.

Bettina Schaeffli: Conceptualization, Writing—review & editing.

Adam Winstral: Writing—review & editing.

Daniel Hunkeler: Conceptualization, Supervision, Writing—review & editing, Project administration, Funding acquisition.

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Data availability The datasets generated and analysed during the current study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest for this manuscript.

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